



Code	Original description	Short description	LED behaviour	Comments	Battery Error Status Register PDO 0x481
0x01	ERROR_NR_OVER_TEMP_LADEN_ZELLEN	Cell over temperature during charging	Solid red	The cell temperature in a module exceeds specified range. Charging on all modules are stopped. Charging is automatically resumed when the temperature of all cells in all modules are within range	Bit 12
0x02	ERROR_NR_UNDER_TEMP_LADEN_ZELLEN	Cell under temperature during charging	Solid red	The cell temperature in a module is below temperature range. Charging on all modules are stopped. Charging is automatically resumed when the temperature of all cells in all modules are within range	Bit 12
0x03	ERROR_NR_OVER_TEMP_LADEN_FET	Charge FET's above temperature limit during charging	Solid red	The Charge FET's temperature in a module exceeds specified range. Charging on all modules are stopped. Charging is automatically resumed when the temperature of all FET's in all modules are within range	Bit 12
0x04	ERROR_NR_OVER_TEMP_ENTLADEN_ZELLEN	Cell over temperature during Discharging	Solid red	The cell temperature exceeds temperature range. Discharging on all modules are stopped. Discharging can only be resumed by charging or Off/On activation if the temperature of all cells in all modules are within range	Bit 13
0x05	ERROR_NR_UNDER_TEMP_ENTLADEN_ZELLEN	Cell under temperature during Discharging	Solid red	The cell temperature below temperature range. Discharging on all modules are stopped. Discharging can only be resumed by charging or Off/On activation if the temperature of all cells in all modules are within range	Bit 13
0x06	ERROR_NR_OVER_TEMP_ENTLADEN_FET	Discharge FET's above temperature limit during discharging	Solid red	The Discharge FET's temperature in a module exceeds specified range. Discharging on all modules are stopped. Discharging can only be resumed by charging or Off/On activation if the temperature of all FET's in all modules are within range	Bit 13
0x07	ERROR_NR_OVER_TEMP_CLAMP	Not used			
0x08	ERROR_NR_OVER_VOLTAGE	Cell over voltage during charge	Solid red	A cell over voltage during charge in any module terminates charge of all modules. Over voltage during charge can only happen if the charger is defect or a non-authorized charger is used. The modules are automatically reset by a discharge detection	Bit 11
0x09	ERROR_NR_UNDER_VOLTAGE	Cell under voltage during discharge	Solid red	When a cell block in a module is depleted below value in Object 0x2104.b or Object 0x2104.c, the complete module is disconnected from the pack. The remaining modules in a pack will continue to operate. The condition is reset by charging to an SoC level of what is defined in 0x2804.3 (Typically: 2%)	Bit 10
0x0a	ERROR_NR_DEEP_LOW_VOLTAGE	Cell voltage too low	Flashing red	If a cell block in a module is deep discharged, this module is permanently disconnected from the pack. The remaining modules in a pack are working normally. The module must be replaced.	Bit 14
0x0b	ERROR_NR_CELL_DISBALANCE	Severe Cell unbalance within a module	Flashing red	If the cell block imbalance inside a module exceeds 200mV, the module will open both FETs and disconnect charging and discharging. If the imbalance drops below 100mV, the failure is reset.	
0x0c	ERROR_NR_AKKU_PACK_SPN_MIN_ERROR	Not used			
0x0d	ERROR_NR_AKKU_PACK_SPN_MAX_ALARM	Overvoltage Pack	Solid red	If the module voltage exceeds 4.5V x number of serial cells, charging is stopped. Charging is resumed if the module voltage drops by 1V. Only the specific module will open the Charge FET.	Bit 5
0x0e	ERROR_NR_AKKU_PACK_FUSED_SPN_DIFF_ERROR	Not used			
0x0f	ERROR_NR_AKKU_PWR_SPN_DIFF_ERROR	Voltage difference between sum of single cells and module voltage	Solid red	The difference of the sum of single cell voltages and the module voltage is too high. Allowed difference is 1V x number of serial cells.	Bit 14
0x10	ERROR_NR_AKKU_PWR_SPN_MIN_ERROR	Measured module voltage too low.	Solid red	The module voltage is less than 2V x number of serial cells. Reset condition: Voltage 1V greater than 2V x number of serial cells.	Bit 14
0x11	ERROR_NR_AKKU_PWR_SPN_MAX_ERROR	Not used			
0x12	ERROR_NR_AKKU_NETZ_SPN_MIN_ERROR	Not used			
0x13	ERROR_NR_AKKU_NETZ_SPN_MAX_ALARM	Not used			
0x14	ERROR_NR_AKKU_REKUPERATIOM_SPN_MAX_ALARM	Not used			
0x15	ERROR_NR_I_CHARGE_SC	Short-circuit current during charging	Solid red	The charge current exceeds 300 Ampere for more than 300 µS. This can only happen if the charger is defect. Charging on all modules is stopped. The module is reset by an Off/On activation or by connecting a load	Bit 3
0x16	ERROR_NR_I_CHARGE_OCC_1	Over current during charging	Solid red	The charge current exceeds 65 Ampere for more than 3 seconds. This can only happen if the charger is defect. Charging on all modules is stopped. The module is reset by an Off/On activation or by connecting a load	Bit 8
0x17	ERROR_NR_I_CHARGE_OCC_2	Over current during charging	Solid red	The charge current exceeds 85 Ampere for more than 50 mS. This can only happen if the charger is defect. Charging on all modules is stopped. The module is reset by an Off/On activation or by connecting a load	Bit 8
0x18	ERROR_NR_I_CHARGE_OCC_3	Not used			
0x19	ERROR_NR_I_DISCHARGE_SC	Short-circuit current during discharging	Solid red	The discharge current exceeds 300 Ampere for more than 300 µS. This can happen if the power terminals are shorted, if the capacitance of the load is too big (Inrush) or if a single module is driving an excessive load. The module is reset by an Off/On activation or by connecting a charger.	Bit 4
0x1a	ERROR_NR_I_DISCHARGE_OCD_1	Over current during discharging	Solid red	The discharge current exceeds 65 Ampere for more than 3 seconds. This happens if the power for a single module is too high and it happens if e.g. 8 of 10 modules in a huge machine is depleted. The remaining 2 modules cannot support the load of the machine. The module is reset by an Off/On activation or by connecting a charger.	Bit 9
0x1b	ERROR_NR_I_DISCHARGE_OCD_2	Over current during discharging	Solid red	The discharge current exceeds 85 Ampere for more than 50 mS. This happens if the power for a single module is too high and it happens if e.g. 9 of 10 modules in a huge machine is depleted. The remaining module cannot support the load of the machine. The module is reset by an Off/On activation or by connecting a charger.	Bit 9
0x1c	ERROR_NR_I_DISCHARGE_OCD_3	Not defined in ComBat-II	N/A	N/A	
0x1d	ERROR_NR_I_AKKU_DIFF_ERROR	Difference of current measurement	Flashing red	The filtered and unfiltered current measurement differs by 10%. Can only happen if there is a HW damage in the current measurement circuitry.	
0x1e	ERROR_NR_SCND_SPN_MIN_ERROR	Not used			
0x1f	ERROR_NR_SCND_SPN_MAX_ERROR	Not used			
0x20	ERROR_NR_SCND_UC_FET_ENABLE	Not used			
0x21	ERROR_NR_SCND_CURRENT_SENSE_EIN	Not used			
0x22	ERROR_NR_SCND_VOLTAGE_SENSE_EIN	Not used			
0x23	ERROR_NR_SCND_TEMP_CELL_SENSE_EIN	Not used			
0x24	ERROR_NR_SCND_TEMP_FET_SENSE_EIN	Not used			
0x25	ERROR_NR_SCND_PYRO_FUSE_EJECT_SENSE_EIN	Not used			
0x26	ERROR_NR_SCND_I_DISCHARGE_FET_ERROR	Not used			
0x27	ERROR_NR_SCND_I_CHARGE_FET_ERROR	Not used			
0x28	ERROR_NR_SCND_VOLTAGE_ERROR	Not used			
0x29	ERROR_NR_ADC_SPN_MIN_SCALE	A value close to the limit of the ADC is measured.	Flashing red	A voltage value greater than 5V is measured which can only happen if there is a damage of the measuring circuitry.	Bit 10
0x2a	ERROR_NR_ADC_SPN_MAX_SCALE	A value close to the limit of the ADC is measured.	Flashing red	A voltage value less than -5V is measured which can only happen if there is a damage of the measuring circuitry.	Bit 11
0x2b	ERROR_NR_ADC_TEMP_ZELLEN_MIN_SCALE	Cell temperature sensor value too high	Flashing red	A cell temperature sensor reports a temperature above +125 °C, which typically means that a temperature sensor is shorted. The module is permanently disconnected from the pack. The remaining modules in a pack are working normally. The module must be replaced.	Bit 14
0x2c	ERROR_NR_ADC_TEMP_ZELLEN_MAX_SCALE	Cell temperature sensor value too low	Flashing red	A cell temperature sensor reports a temperature below -40 °C, which typically means that a temperature sensor is disconnected. The module is permanently disconnected from the pack. The remaining modules in a pack are working normally. The module must be replaced.	
0x2d	ERROR_NR_ADC_TEMP_FET_MIN_SCALE	FET temperature sensor value too high	Flashing red	A FET temperature sensor reports a temperature above +125 °C, which typically means that a temperature sensor is shorted. The module is permanently disconnected from the pack. The remaining modules in a pack are working normally. The module must be replaced.	Bit 14
0x2e	ERROR_NR_ADC_TEMP_FET_MAX_SCALE	FET temperature sensor value too low	Flashing red	A FET temperature sensor reports a temperature below -40 °C, which typically means that a temperature sensor is disconnected. The module is permanently disconnected from the pack. The remaining modules in a pack are working normally. The module must be replaced.	
0x2f	ERROR_NR_ADC_TEMP_CLAMP_MIN_SCALE	Not used			
0x30	ERROR_NR_ADC_TEMP_CLAMP_MAX_SCALE	Not used			
0x31	ERROR_NR_ADC_I_CHARGE_MIN_SCALE	Not used			
0x32	ERROR_NR_ADC_I_CHARGE_MAX_SCALE	A value close to the limit of the ADC is measured.	Flashing red	A current value greater than +400 A is measured. NOTE: The delay is longer than the short circuit protection.	Bit 8
0x33	ERROR_NR_ADC_I_DISCHARGE_MIN_SCALE	Not used			
0x34	ERROR_NR_ADC_I_DISCHARGE_MAX_SCALE	A value close to the limit of the ADC is measured.	Flashing red	A current value less than -600A is measured. NOTE: The delay is longer than the short circuit protection.	Bit 9
0x35	ERROR_NR_I_DISCHARGE_FET_ERROR	Discharge FET is shorted	Flashing red	Discharging cannot be stopped but recharge is not possible. The module must be replaced	Bit 6
0x36	ERROR_NR_I_CHARGE_FET_ERROR	Charge FET is shorted	Flashing red	Discharging and Charging is not possible. The module must be replaced	Bit 7
0x37	ERROR_NR_I_DISCHARGE_CHARGE_FET_ERROR	Not used			
0x38	ERROR_NR_TEMP_DISCHARGE_ERROR_LOCK	Discharge Error Lock	Solid red	This flag is set in combination with Error 0x04 / 0x05 / 0x06.	Bit 15
0x39	ERROR_NR_TEMP_CHARGE_ERROR_LOCK	Charge Error Lock	Solid red	This flag is set in combination with Error 0x01 / 0x02 / 0x03.	
0x3a	ERROR_NR_OVER_CHARGE_CURRENT_ALARM_RECUPERATION	Overcharge current due to recuperation	Solid red	Pack over current during discharge can only happen due to recuperation. The modules are automatically reset by a discharge detection	Bit 2
0x3b	ERROR_NR_OVER_CHARGE_CELL_VOLTAGE_ALARM_RECUPERATION	Overcharge cell voltage due to recuperation	Solid red	Cell over voltage during discharge can only happen due to recuperation. The modules are automatically reset by a discharge detection	Bit 2
0x3c	ERROR_NR_24V_SPN_MIN_ERROR	24-volt supply below limit	Flashing red	24V supply of the BMS is lower than 16V.	Bit 15
0x3d	ERROR_NR_24V_SPN_MAX_ERROR	24-volt supply above limit	Flashing red	24V supply of the BMS is higher than 26V.	Bit 15
0x3e	ERROR_NR_CAN_NETWORK_NOT_CONF_NODE_ID	Node-ID are not assigned	Flashing Orange	The module cannot communicate on CAN and charging is not possible	
0x3f	ERROR_NR_CAN_NETWORK_DOUBLE_NODE_ID	NodeID conflict	Flashing Orange	The module cannot communicate with other Nodes on CAN and charging is not possible	
0x40	ERROR_NR_PARAMETER_CONFIGURATION_ERROR	Parameter outside range	Flashing red	The version of the module is identified from the SW via an external ID resistors. If the ID resistor coding doesn't fit to the programming of the specific variant (24V / 36V / 48V), this failure is set.	Bit 15

Pushed (Always available)

Pulled (Available upon request)